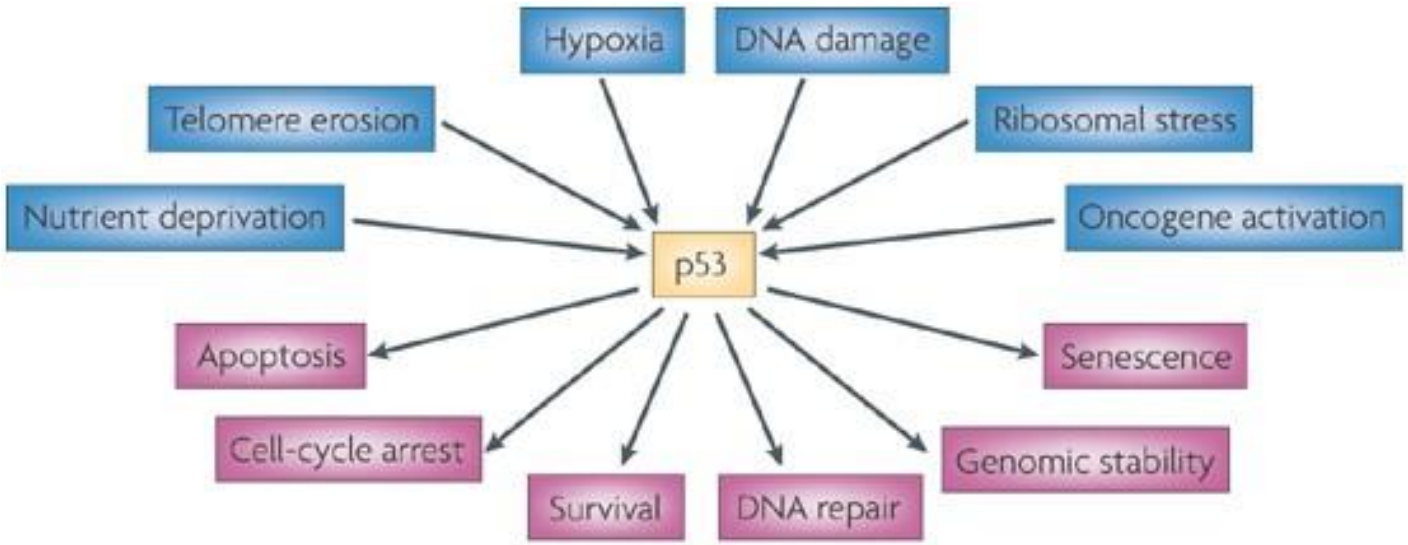
TP53 MUTATIONS THAT ARE ASSOCIATED WITH AN INCREASED COPY NUMBER VARIATION

Cancer Bioinformatics - Final Project

Ornit Bhonkar

March 19, 2026

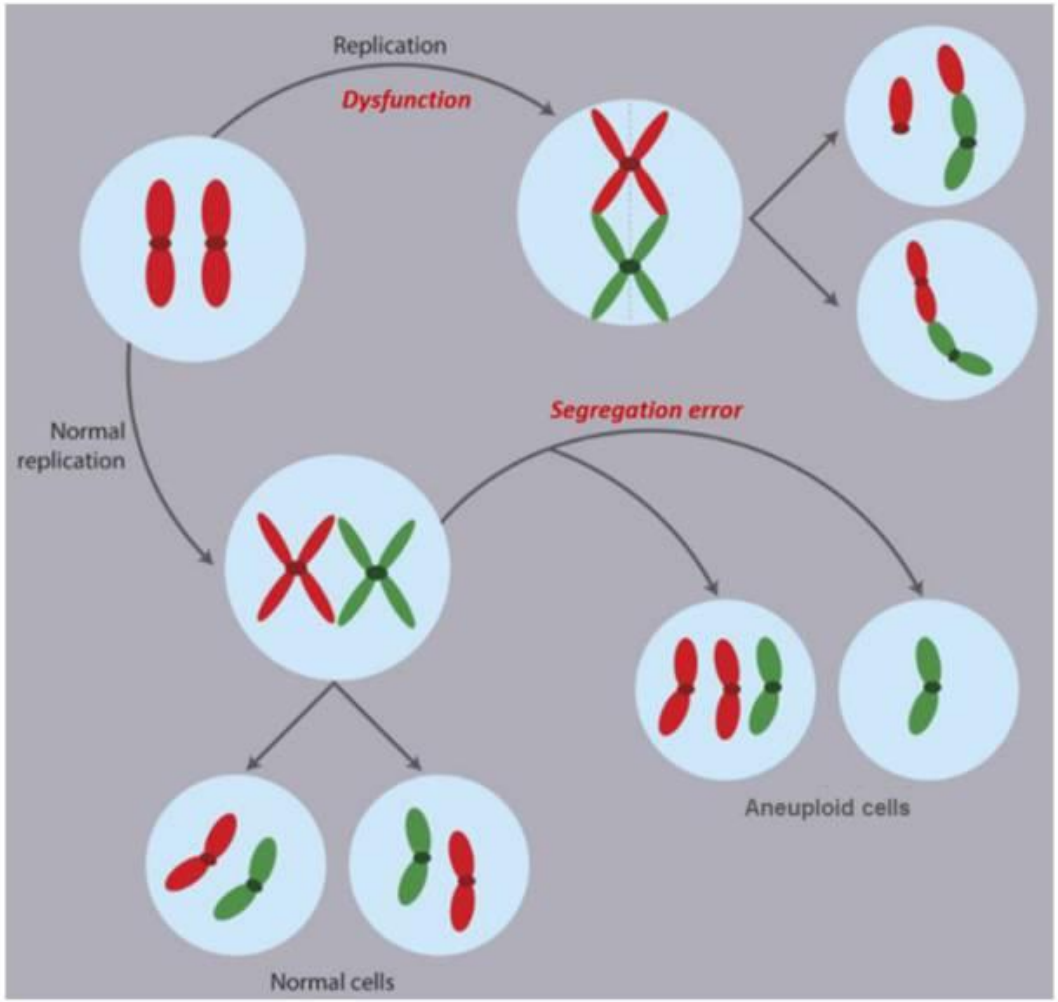
TP53



- TP53 acts as the cellular "superhero" responsible for maintaining genomic integrity. It functions as a stress sensor that responds to DNA damage.
- When TP53 is functional, it induces DNA repairs or apoptosis. When broken, errors accumulate.
- TP53 mutations are not binary, and they exist on a spectrum of phenotypic impact.

COPY NUMBER VARIATION

- A stretch of DNA where the number of copies differs from the normal two copies.
  - More copies → more RNA → more protein
  - Fewer copies → less protein
  - Example: Oncogene amplification → uncontrolled growth

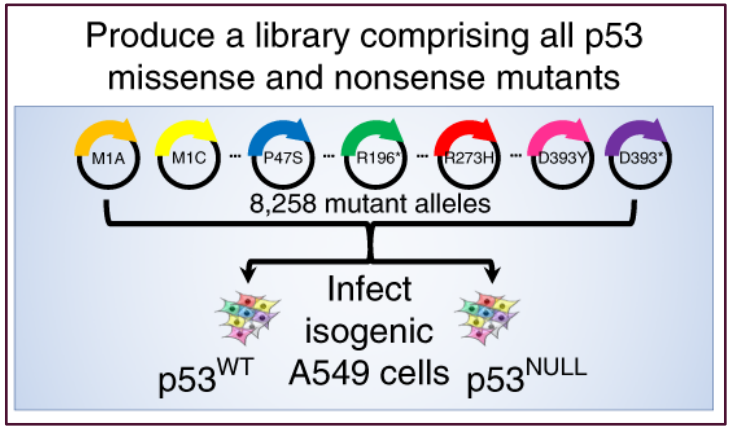


EXPERIMENTAL FUNCTIONAL SCORE

Screen 8,258 TP53 Variants

Run 3 competitive growth screens in isogenic p53WT and p53NULL cells treated with nutlin-3 or etoposide. Collect read counts to quantify allele representation.

1



Combined Phenotype Score (EFS)
Low = Loss of Function
High = WT-like Activity

Quantify Allele Enrichment

Compute $\log_2$ fold-change for each allele relative to the early timepoint. Standardize across alleles to produce a Z-score for each of the three conditions.

2

3

EFS = [Z(WT+nutlin) + Z(NULL+nutlin) − Z(NULL+etop)] / 3

Calculate Functional Score

Integrate the three Z-scores using the Combined Phenotype Score formula. This single value per allele - the EFS - reflects the degree to which each TP53 variant retains or loses normal p53 function.

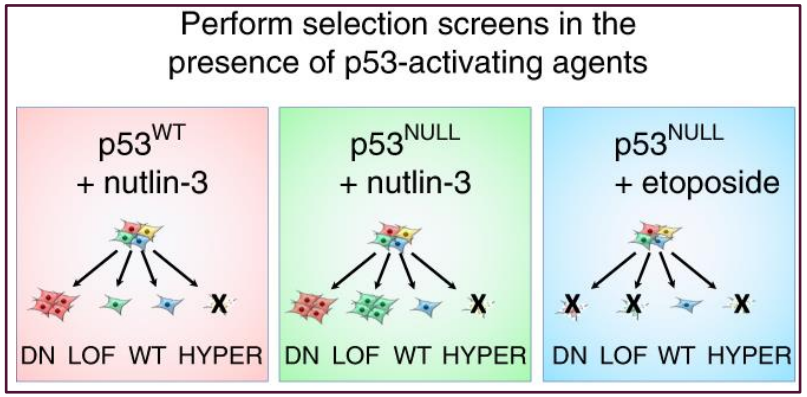
Giacomelli, A.O., Yang, X., Lintner, R.E. *et al.* Mutational processes shape the landscape of *TP53* mutations in human cancer. *Nat Genet* 50, 1381–1387 (2018).

EXPERIMENTAL FUNCTIONAL SCORE

Screen 8,258 TP53 Variants

Run 3 competitive growth screens in isogenic p53WT and p53NULL cells treated with nutlin-3 or etoposide. Collect read counts to quantify allele representation.

1



Combined Phenotype Score (EFS)
Low = Loss of Function
High = WT-like Activity

Quantify Allele Enrichment

Compute $\log_2$ fold-change for each allele relative to the early timepoint. Standardize across alleles to produce a Z-score for each of the three conditions.

2

3

EFS = [Z(WT+nutlin) + Z(NULL+nutlin) − Z(NULL+etop)] / 3

Calculate Functional Score

Integrate the three Z-scores using the Combined Phenotype Score formula. This single value per allele - the EFS - reflects the degree to which each TP53 variant retains or loses normal p53 function.

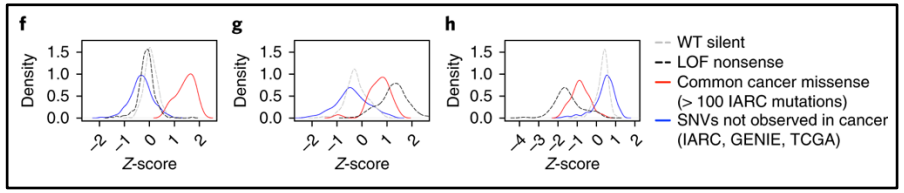
Giacomelli, A.O., Yang, X., Lintner, R.E. *et al.* Mutational processes shape the landscape of *TP53* mutations in human cancer. *Nat Genet* 50, 1381–1387 (2018).

EXPERIMENTAL FUNCTIONAL SCORE

Screen 8,258 TP53 Variants

Run 3 competitive growth screens in isogenic p53WT and p53NULL cells treated with nutlin-3 or etoposide. Collect read counts to quantify allele representation.

1



Combined Phenotype Score (EFS)
Low = Loss of Function
High = WT-like Activity

Quantify Allele Enrichment

Compute $\log_2$ fold-change for each allele relative to the early timepoint. Standardize across alleles to produce a Z-score for each of the three conditions.

2

3

EFS = [Z(WT+nutlin) + Z(NULL+nutlin) − Z(NULL+etop)] / 3

Calculate Functional Score

Integrate the three Z-scores using the Combined Phenotype Score formula. This single value per allele - the EFS - reflects the degree to which each TP53 variant retains or loses normal p53 function.

Giacomelli, A.O., Yang, X., Lintner, R.E. *et al.* Mutational processes shape the landscape of *TP53* mutations in human cancer. *Nat Genet* 50, 1381–1387 (2018).

HYPOTHESIS

Patients with TP53 mutations that have the lowest experimental function scores (EFS) will show the highest Fraction Genome Altered (FGA)

RESEARCH QUESTIONS

Is there a correlation between the experimental TP53 EFS (Giacomelli et al.) and patient FGA (GENIE)?

Do "Hotspot" mutations drive higher genomic instability than rare mutations?

Does this correlation hold true across all cancer types or is it specific to certain tissues?

DATA SOURCES: INTEGRATING PHENOTYPE & GENOTYPE

Primary Analysis

- AACR GENIE v19.0 (~271,000 samples, 100,418 with FGA).
- Role: Real-world patient data - the phenotype
- Metric: FRACTION_GENOME_ALTERED (FGA), a quantitative measure of chromosomal instability
- 35,357 TP53-mutant
- 65,061 TP53-WT

- Giacomelli et al., Nature Genetics, 2018 (25,217 variants, 8,274 unique)
- Role: Experimental "fitness" scores for every possible TP53 mutation - the genotype
- Metric: Experimental Functional Score (derived from cell line fitness screens) quantifies exactly how well each specific TP53 mutant can suppress tumor growth in a controlled lab setting

Goal: to show that the physical structure of the genome is breaking down

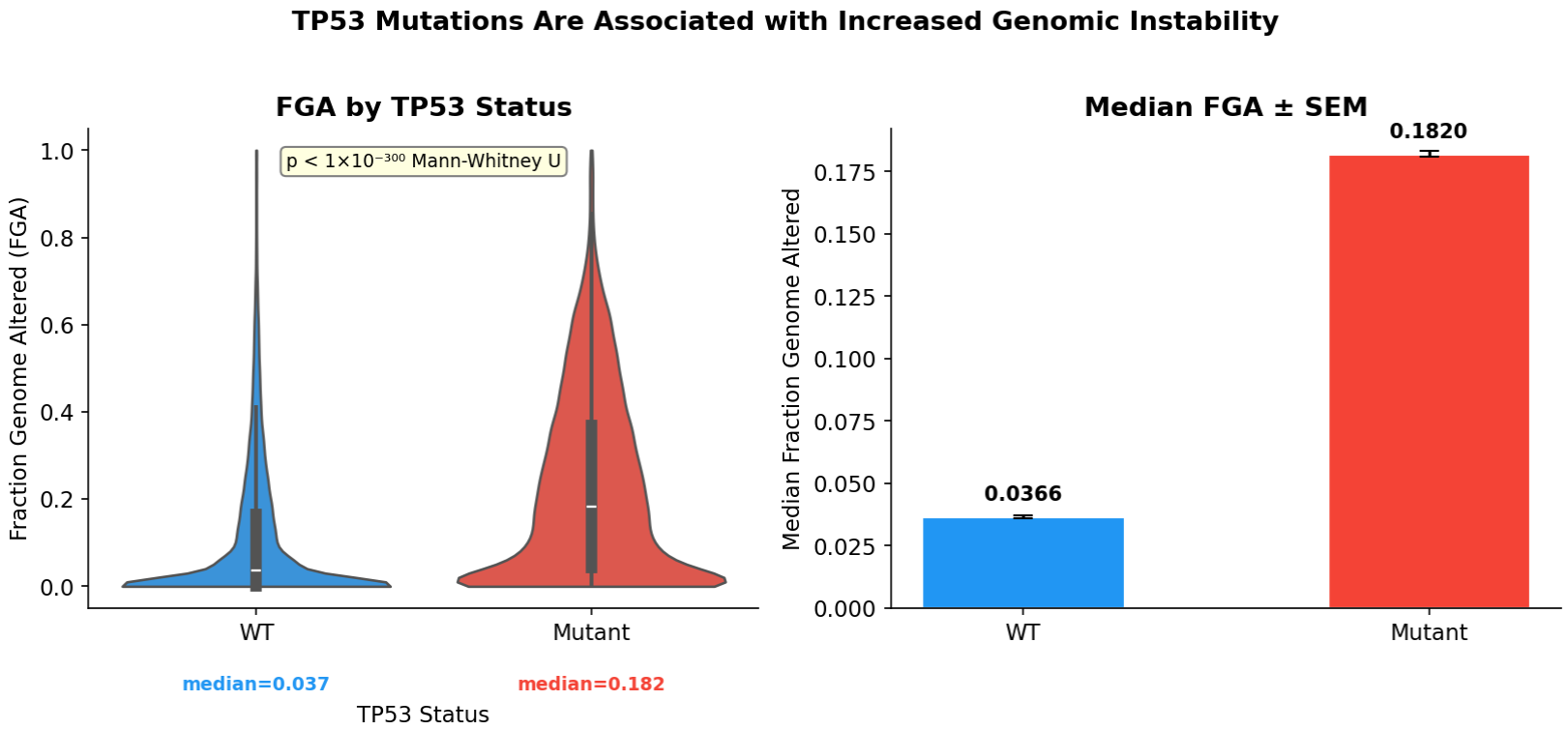
DATA SOURCES: INTEGRATING PHENOTYPE & GENOTYPE

Secondary Analysis

- MSK-IMPACT (53,292 samples, 505 cancer genes).
- Role: Genome-wide CNA data to identify which pathways are altered when TP53 breaks
- Compared 800 high-FGA TP53-mutant vs 800 low-FGA TP53-WT

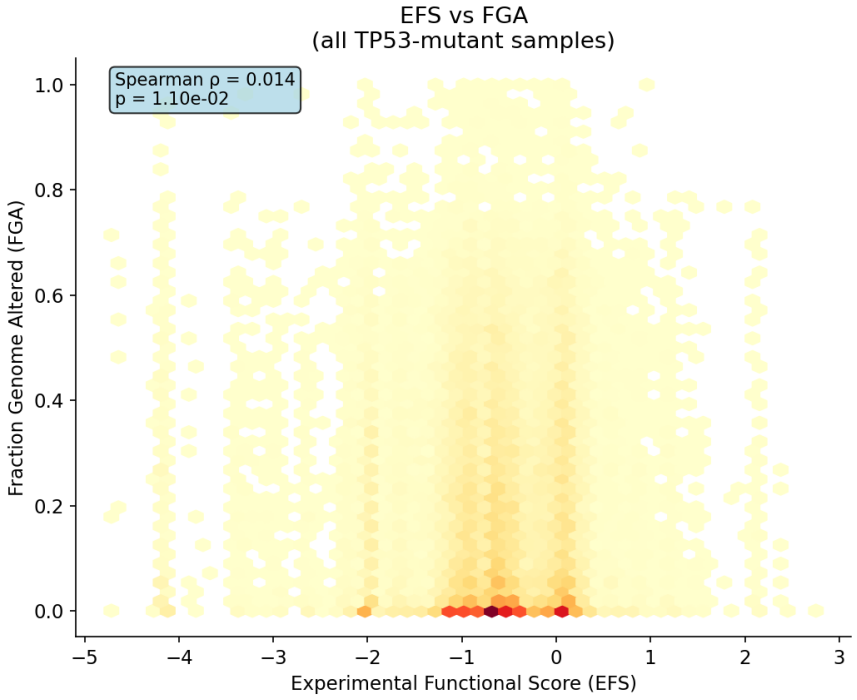
Goal: to conclude what are the major pathways are triggered

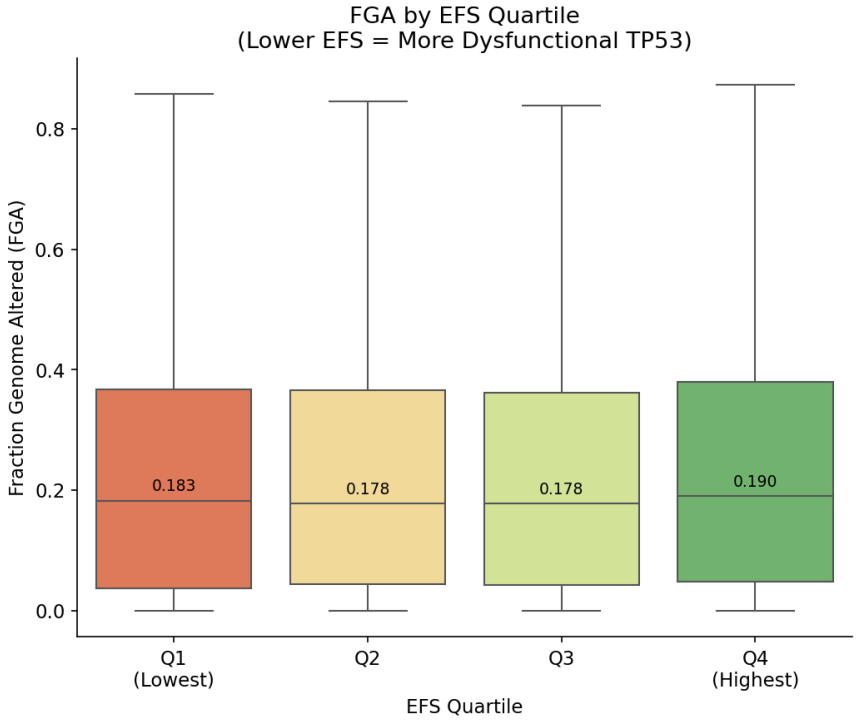
DO TP53 MUTATIONS CORRELATE WITH HIGHER GENOMIC INSTABILITY (FGA)?



TP53 mutant tumors have significantly higher FGA than WT (median 0.182 vs 0.037, $p < 1\times10^{-300}$)

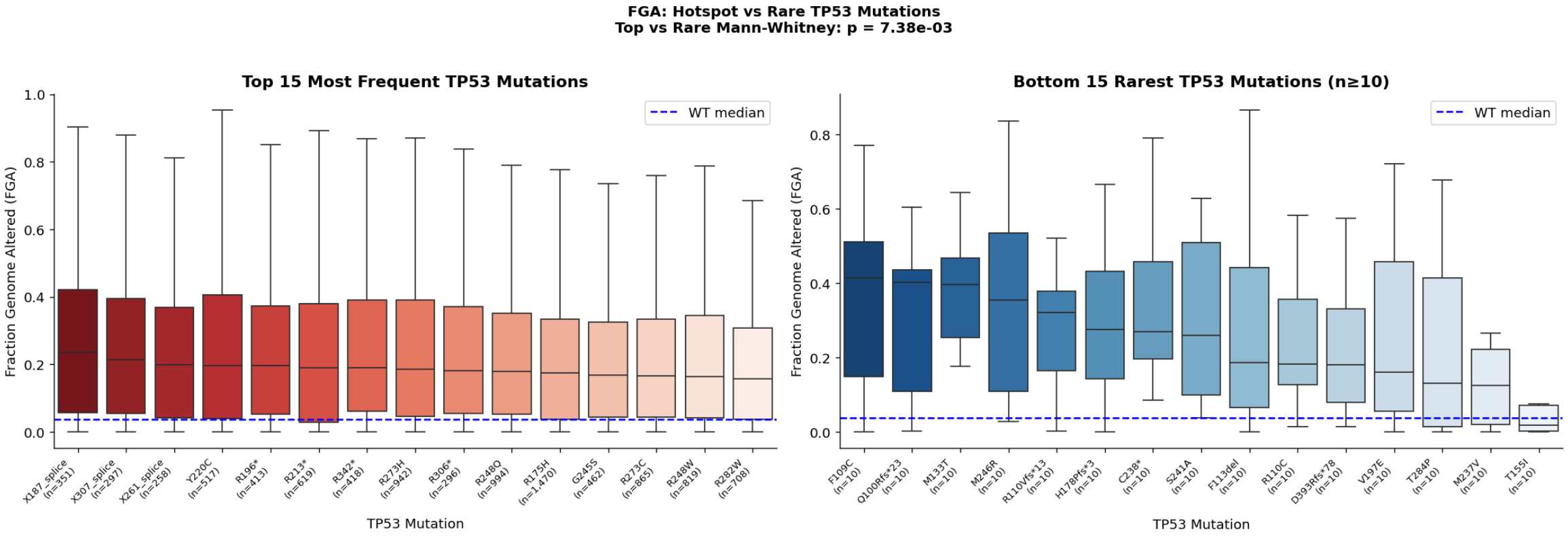
IS THERE A CORRELATION BETWEEN EFS AND FGA?





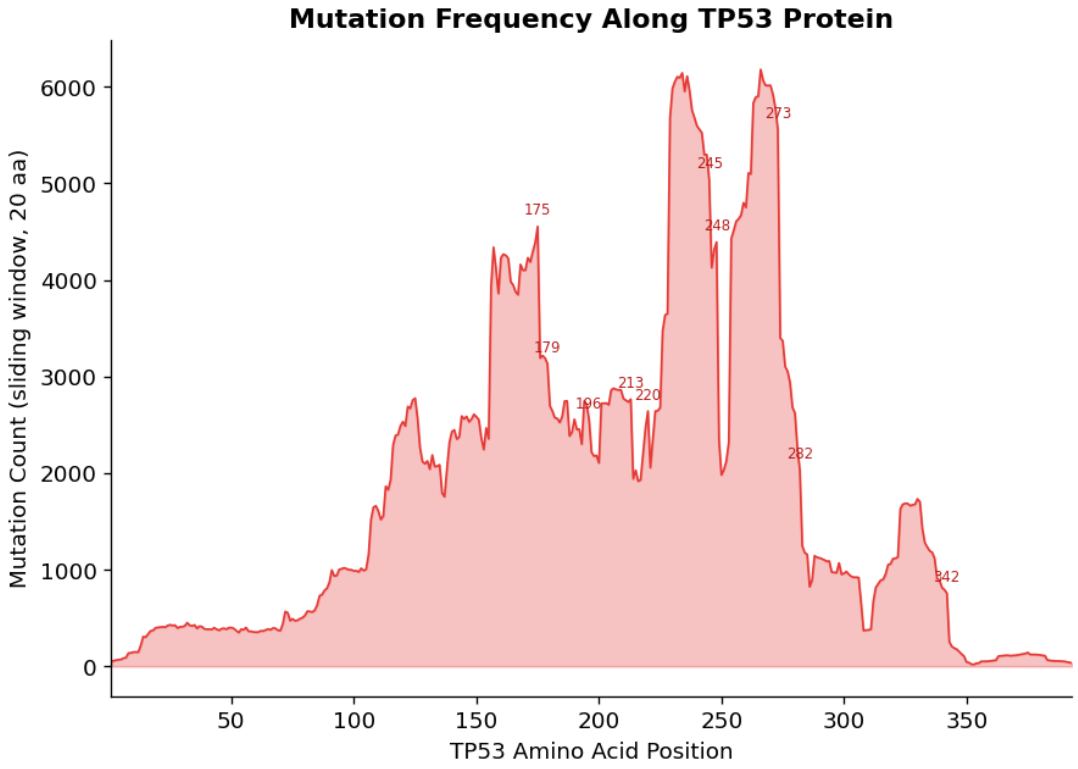
Weak correlation. The hypothesis partially holds but EFS alone doesn't predict FGA well

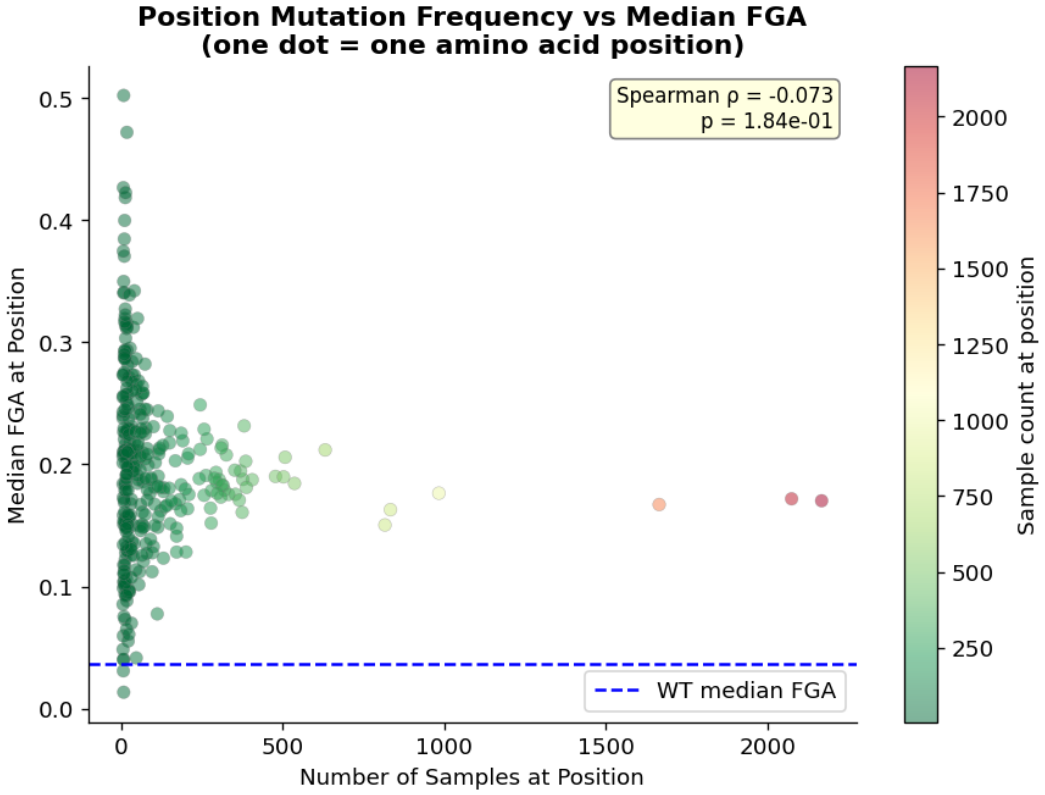
DO "HOTSPOT" MUTATIONS DRIVE HIGHER GENOMIC INSTABILITY THAN RARE MUTATIONS?



Rare mutations have higher median FGA. ”Hotspot” mutations have lower FGA median than rare mutations

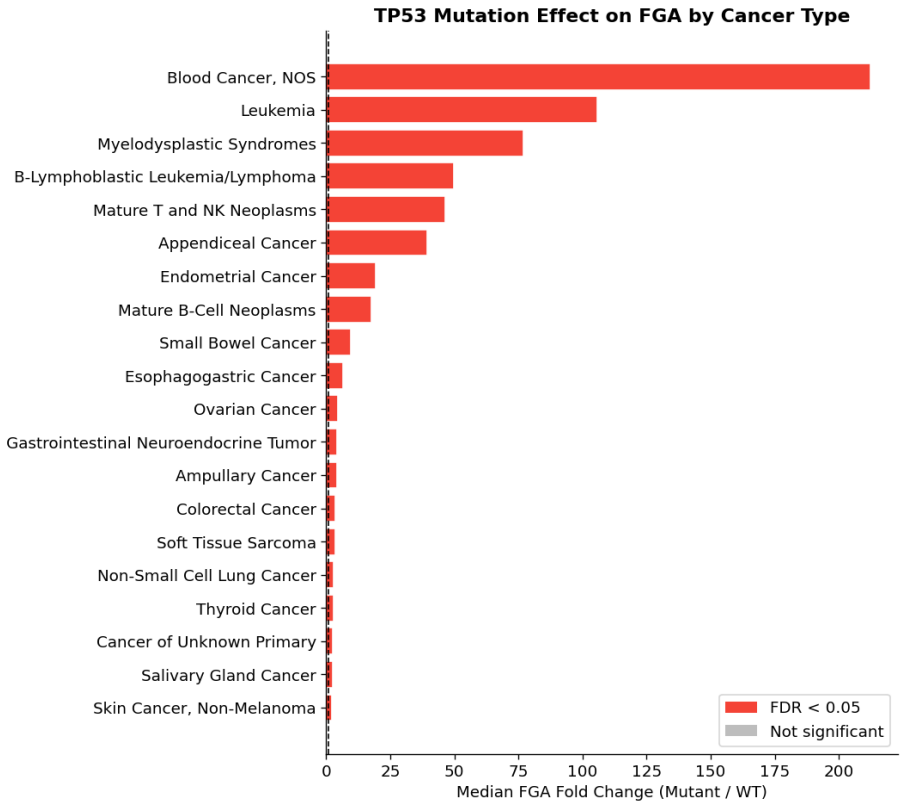
DO "HOTSPOT" MUTATIONS DRIVE HIGHER GENOMIC INSTABILITY THAN RARE MUTATIONS?

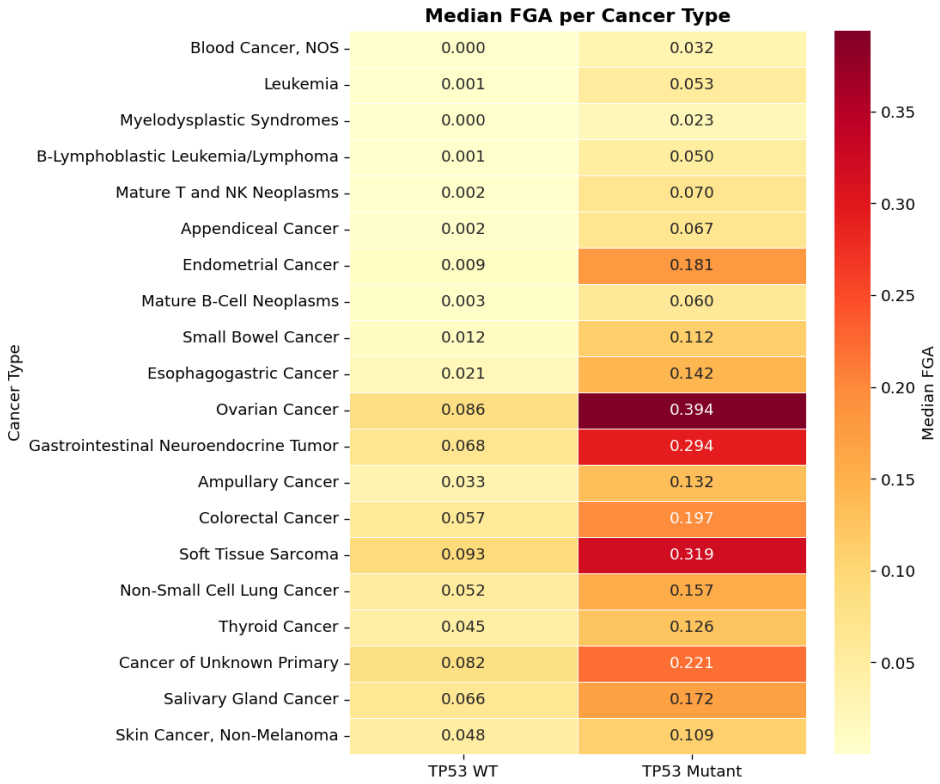




The loss of TP53 function itself drives genomic instability uniformly.

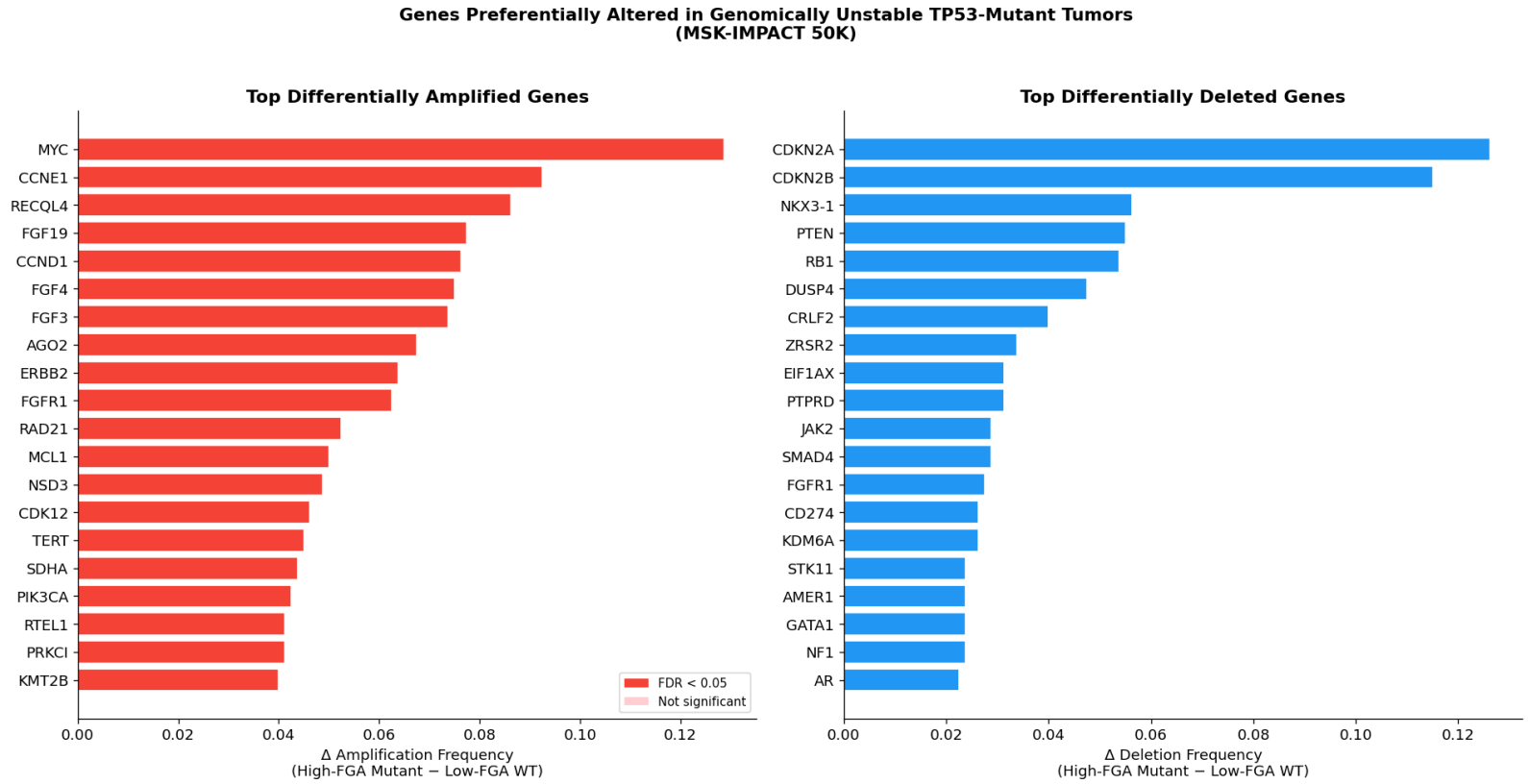
IS THIS CANCER-TYPE SPECIFIC?





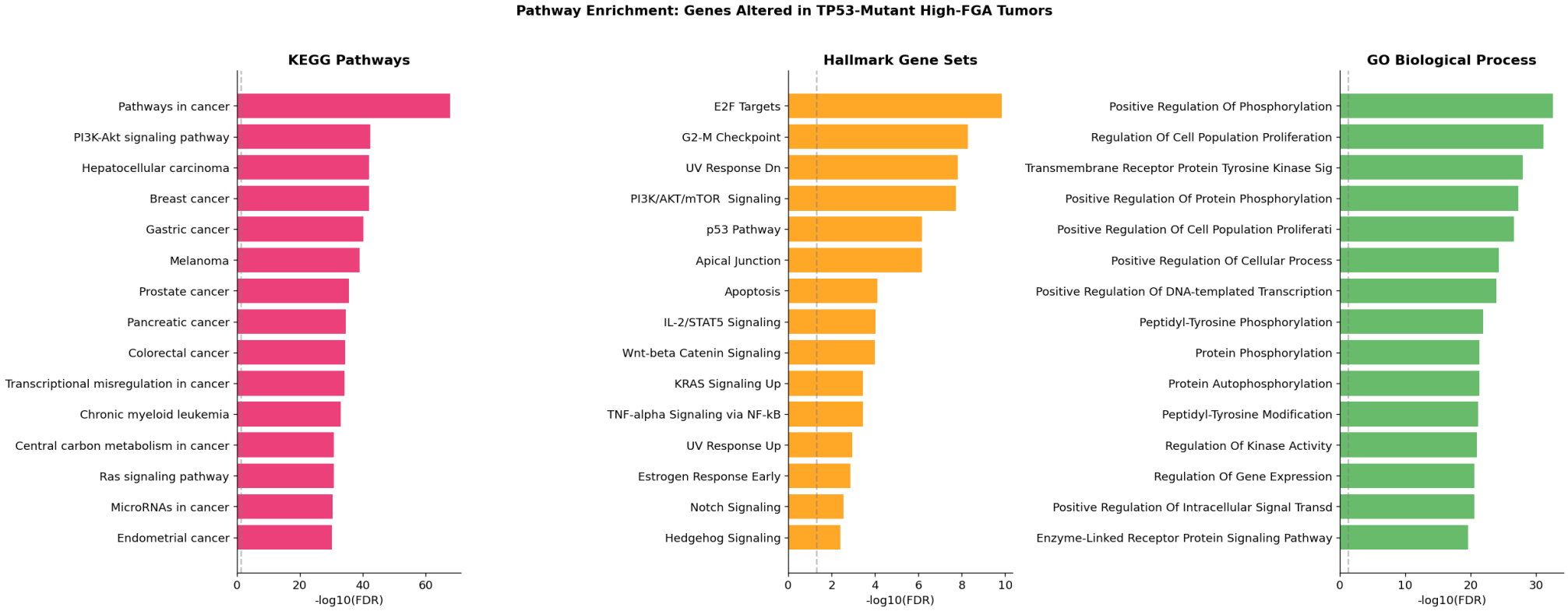
All top absolute FGA cancers are solid tumors (Ovarian Cancer, Soft Tissue Sarcoma, and Colorectal Cancer). The blood cancers have the highest fold change but tiny absolute FGA values (~0.03–0.07) because their WT baseline is essentially zero.

WHICH KNOWN CANCER GENES CO-OCCUR WITH TP53 LOSS AND HIGH FGA?



Amplified genes (left panel): oncogenes being turned ON
Deleted genes (right panel): tumor suppressors being turned OFF

WHICH PATHWAYS ARE TRIGGERED?



When TP53 breaks and FGA is high, cell cycle control (E2F, G2-M) and growth signaling (PI3K/AKT/mTOR) pathways are the main downstream consequences

BINARY EFFECT VS. CONTINUOUS EFFECT

- The binary effect (mutant vs WT) is stronger than the continuous effect (EFS vs FGA).
- This suggests a threshold model: once TP53 loses function, genomic chaos follows, but the specific degree of functional loss matters less than its presence
- Other tumor-intrinsic and tissue-specific factors play a larger role

THE HOTSPOT PARADOX

- Hotspot mutations dominate in tumors not because they're functionally the worst (Giacomelli showed this), but because their DNA sequences are highly mutable. Yet they still drive genomic instability, consistent with the threshold model.

LIMITATIONS

- EFS derived from one lung cell line. That may not generalize across tissues
- FGA is a bulk measure, and it doesn't capture which specific chromosomal regions break

CONCLUSION

TP53 loss drived genomic instability as a class effect, which in turn selectively amplifies oncogenic growth pathways, which explains why TP53 mutation is such a universal and aggressive cancer event.

THANK YOU!